A uniqueness theorem for the container derivative

∂ is the only nontrivial representable first-order tangent structure on the linear layer $\mathsf{Lin}(\mathsf{SPoly})$ of finite-support polynomial functors

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4 October 2026 (revised 7 October 2026: ambient category $\mathsf{Lin}(\mathsf{SPoly}) \simeq$ finite free $\mathbb{N}$-modules; solidity restated as invertibility of the vertical lift; finite-rank sharpness withdrawn — uniqueness holds at every rank. Revised 8 October 2026 per second-pass referee report: the rank-free exclusion (Theorem 13) and the solidity of the two surviving parts (Theorem 18) are now proved in full within the note; two scope overclaims corrected. Revised 10 October 2026 per third-pass referee report: scope of "only nontrivial" pinned to representable Weil base change, footnote 1 reworded, and solidity disambiguated into abstract and vertical-lift senses — all wording only, except that Proposition 12 was restated as requested (title now "abstractly solid", dropping the clause "hence an isomorphism ℓ_M"))

Built from commit `6089268` of `github.com/scotmacbeth/work-in-progress`.
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Abstract

A companion note showed that the Abbott–Altenkirch–Ghani–McBride one-hole derivative ∂ is the differential combinator of a Cartesian differential category SPoly of polynomial functors, with the dual-number substitution $p \mapsto p(y + \varepsilon v) \bmod \varepsilon^2$ computing its combinator; that settles *existence*. This note settles *uniqueness*. The right setting is the subcategory $\mathsf{Lin}(\mathsf{SPoly})$ of *linear* maps — morphisms given by $\mathbb{N}$-matrices, composition the matrix product — equivalent to the finite-rank part of the category $\mathsf{FreeMod}_{\mathbb{N}}$ of free $\mathbb{N}$-modules. On this subcategory the tangent structure is *base change* $T = (-) \otimes_{\mathbb{N}} D$ by an augmented square-zero $\mathbb{N}$-algebra (a Weil algebra over $\mathbb{N}$), the dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2$ being the first instance; this is the module layer of Lanfranchi–Lemay's representable tangent structures, now over the rig $\mathbb{N}$. We prove that among representable first-order tangent structures of this form there are *exactly two*, the trivial structure $T = \mathrm{id}$ and the container derivative ∂. The proof has two halves. The first is a rank rigidity: an infinitesimal object $D = y \oplus M$ has a *solid* infinitesimal part when the vertical-lift component $\ell_M \colon M \to M \otimes M$ is invertible, and over $\mathbb{N}$ a linear module is automatically free of a well-defined rank, so solidity forces $|S|^2 = |S|$, whose only finite solutions are 0 and 1. The second is the converse: every such tangent structure has a solid infinitesimal part. We prove this rig-internally from the universality of the vertical lift, replacing Lanfranchi–Lemay's ring-theoretic short-exact-sequence step — which needs subtraction — by the observation that the vertical bundle is a *direct sum* of its surviving summands, so its rank is computed by addition: free $\mathbb{N}$-modules are closed under direct sum and rank is additive over it ($\mathrm{rank}(A \oplus B) = \mathrm{rank}\, A + \mathrm{rank}\, B$), with no subtraction available to undo it. The port is thus not only faithful but cleaner: no

*Kodamai. Source and Lean certification: `github.com/scotmacbeth/work-in-progress`. Earlier versions (commit `57bac18`) mis-stated the ambient category, and the revision of 5 October mis-stated solidity and the sharpness of the finite-rank hypothesis; both are corrected here, see §2, §5.

principal-ideal hypothesis and no subtraction. Crucially, on the linear layer base change $\otimes D$ coincides with container substitution $\triangleleft$ (and with the Dirichlet tensor), so the classification is at once a statement about Weil base change and about the container derivative — but it is *not* a statement about $\triangleleft$-substitution on all of Poly, where there is no $\varepsilon^2 = 0$ truncation ($y^2 \triangleleft 2y = 4y^2$) and only $T = \mathrm{id}$ survives. The uniqueness holds for a free infinitesimal part of *any* rank. Finiteness is *not* load-bearing: although $M = \mathbb{N} \cdot y$ is *abstractly* solid, because $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ supplies an abstract isomorphism $M \otimes M \cong M$, abstract solidity is strictly weaker than underlying a tangent structure at infinite rank — the flip and lift-coassociativity axioms force the vertical lift to be the diagonal, which is surjective only in rank ≤ 1, so $\mathbb{N} \cdot y$ is *not* a tangent structure (an earlier draft's "finiteness is sharp" claim, which measured solidity by an abstract isomorphism, is withdrawn). The uniqueness answers, and sharpens, a question of Cockett (2012).

1 Introduction

From existence to uniqueness. In a companion note [11] we showed that the category SPoly of finite-support polynomial functors is a Cartesian differential category whose differential combinator is the Abbott–Altenkirch–Ghani–McBride one-hole-context derivative ∂ [1], and that its combinator is computed by the oldest trick in differential algebra: substitute a dual number into a *map* and read off the ε-coefficient,

$$\partial[f](x, v) \;=\; \varepsilon\text{-coefficient of } f(x + \varepsilon v) \bmod \varepsilon^2 \;=\; v \cdot \partial f.$$

That note answers an *existence* question — there is a tangent structure whose vertical part is ∂. The natural next question is *uniqueness*. This ∂-structure is one tangent structure; the identity functor $T = \mathrm{id}$ is another (the trivial one). Are there others? Could there be an exotic first-order tangent bundle that is neither the derivative nor trivial? This note proves there cannot, among representable (Weil base-change) structures on the linear layer.

First we must say precisely in which category the question is asked — and here the right setting is not Poly at large but its *linear* subcategory. Let $\mathsf{Lin}(\mathsf{SPoly})$ be the subcategory of SPoly whose morphisms are the *linear* maps $n \to m$ — those given by an $\mathbb{N}$-matrix, with composition the matrix product — which is equivalent to the category $\mathsf{FreeMod}_{\mathbb{N}}$ of finite free $\mathbb{N}$-modules. On $\mathsf{Lin}(\mathsf{SPoly})$ the tangent structure is *base change* $T = (-) \otimes_{\mathbb{N}} D$ by an augmented square-zero $\mathbb{N}$-algebra D (a Weil algebra over $\mathbb{N}$, the dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2$ being the first instance); this is precisely the module layer on which Lanfranchi–Lemay's representable tangent structures live, now over the rig $\mathbb{N}$ rather than a ring. Two cautions fix the setting. First, base change is *not* the substitution formula: the ring map $\mathbb{N}[y] \to W[y]$ sends $y^2 \mapsto y^2$, whereas the dual-number *substitution* $y \mapsto y + \varepsilon v$ reads off $y^2 + 2\varepsilon v y$; the two agree only on *linear* arguments, and a non-linear polynomial is in any case neither an object nor a morphism of $\mathsf{FreeMod}_{\mathbb{N}}$. The substitution formula is how the companion note computes the combinator ∂ at the level of SPoly *maps* [11]; the tangent *functor* on modules is base change. Second, it is *tempting*, but wrong, to read T as right-substitution $(-) \triangleleft W$ in the full substitution-monoidal category $(\mathsf{Poly}, \triangleleft, y)$: full $\triangleleft$ has no $\varepsilon^2 = 0$ truncation ($y^2 \triangleleft 2y = (2y)^2 = 4y^2$, not $3y^2$; in general $y^n \triangleleft W = W^n$ is multiplicative), so on all of Poly the functor $(-) \triangleleft W$ forces the trivial structure alone. The derivative survives precisely because the truncation — the Weil algebra — is present. What rescues the container reading is a coincidence (Lemma 7): on the linear layer $S \cdot y$, base change $\otimes D$, substitution $\triangleleft$, and the Dirichlet tensor all agree, and it is on this layer — the finite-rank part of $\mathsf{FreeMod}_{\mathbb{N}}$, namely $\mathsf{Lin}(\mathsf{SPoly})$ — that the container-derivative reading of the classification lives.

Theorem 1 (Uniqueness)**.** *Among representable first-order tangent structures $T = (-) \otimes D$ on* $\mathsf{FreeMod}_{\mathbb{N}}$ *— base change by an augmented square-zero $\mathbb{N}$-algebra $D = y \oplus M$ with M a free $\mathbb{N}$-module of* any *rank — there are exactly two: the trivial structure ($D = y$, $M = 0$, $T = \mathrm{id}$) and the AAGM dual-number structure ($D = W = \mathbb{N}[\varepsilon]/\varepsilon^2$, $M = \langle \varepsilon v \rangle$, combinator ∂). Restricted to the finite-rank part* $\mathsf{Lin}(\mathsf{SPoly})$ *— the linear layer of* SPoly*, where $\otimes D$ and $\triangleleft$ coincide — this says the container derivative is the only nontrivial representable first-order tangent structure.*

The question is Cockett's. That polynomial functors should carry a differential structure is not a new hope. In an unpublished talk at FMCS 2012, *Can you Differentiate a Polynomial?*, Cockett [6] conjectured exactly this — that polynomial functors form a Cartesian differential, hence a tangent, category — fourteen years before the companion note proved it. But the 2012 slides are candid about how much was conjecture: the differentiation there is defined only on "separable" polynomials, closure under composition is asserted but not proved, and the chain rule is flagged as the hard open point ("*of course, need to prove this is a differential!! (chain rule is already challenging...)*"; "*Is this all completely sorted out? Absolutely not!*"). No uniqueness statement appears, nor could it: without the structure in hand there is nothing to count. The companion note supplied the structure; the present note supplies the count, and so closes the arc Cockett opened. (We flag one divergence in §9: combinatorial species, the setting nearest Cockett's, are most naturally a tangent *bicategory* [10], structurally different from the SPoly one-category we work in.)

The template, and why it does not simply transfer. For commutative algebra the uniqueness question has a sharp answer. Lanfranchi and Lemay [9] classify the *representable* tangent structures on the opposite of the category of modules (equivalently, affine schemes): a representable tangent structure is a tensoring $T = D \otimes (-)$ by an *infinitesimal object* $D = R \oplus M$, and when the base ring R is a principal ideal domain there are *exactly two* — the trivial one ($M = 0$) and the Kähler/dual-number one ($M = R$). Their argument has two load-bearing uses of ring structure. The *rigidity* half (a solid infinitesimal part has rank n with $R^{n^2} \cong R^n$, forcing $n^2 = n$, hence $n \in \{0, 1\}$; their Theorem 4.17, Corollary 4.18) needs R to be a PID, so that the infinitesimal module is free with a well-defined rank. The *converse* half (a tangent structure's infinitesimal part *is* solid; their Theorem 4.8, via the universality of the vertical lift) computes a vertical bundle by a short exact sequence — i.e. by *subtraction* of ranks. Neither transfers to polynomial functors verbatim: the base there is the rig $\mathbb{N}$, which is not a PID and has no subtraction.

What we prove, and the one new idea. We prove Theorem 1 by porting *both* halves of Lanfranchi–Lemay to the Weil base-change setting over $\mathbb{N}$ — the tangent functor $T = (-) \otimes W$ on $\mathsf{FreeMod}_{\mathbb{N}}$, whose finite-rank part is $\mathsf{Lin}(\mathsf{SPoly})$ — and in both the port turns out *cleaner* than its template. Throughout, the objects that carry the classification are *linear* ($S \cdot y$), the layer on which base change $\otimes W$, substitution $\triangleleft$ and the Dirichlet tensor coincide (Lemma 7); this is what licenses the container-derivative reading of the result.

- *Rigidity* (§4). Inside $\mathsf{FreeMod}_{\mathbb{N}}$ the infinitesimal part M is a free module with a well-defined rank, so freeness is *automatic*, with no PID hypothesis. Solidity (invertibility of the vertical lift ℓ_M) gives an abstract isomorphism $M \otimes M \cong M$, which reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1: the finite solid parts are exactly $\{0, y\}$.
- *Converse* (§6). Every such tangent structure has a solid infinitesimal part. Here is the one new idea. The place Lanfranchi–Lemay use subtraction is the computation of the vertical

bundle fibre P' by exactness. Over $\mathbb{N}$ that fibre is instead a *direct sum* of the surviving summands,

$$P' \;=\; y \;\oplus\; \varepsilon_{\text{out}} \;\oplus\; (M \lhd M),$$

so its rank is a *sum*, $1 + n + n^2$, computed by addition and never as a difference. The universality of the vertical lift (Cockett–Cruttwell [7]) identifies the second tangent bundle T_2 with the vertical bundle V; evaluating this functor isomorphism at y gives $D_2 \cong P'$, i.e. $1+2n = 1+n+n^2$, and the cancellative rig $\mathbb{N}$ (no zero divisors, *no subtraction*) forces $n^2 = n$, hence $n \in \{0, 1\}$.

- *Rank-free exclusion* (§5). The rank count above needs finiteness, since $\kappa^2 = \kappa$ for infinite κ. But the vertical lift of a tangent structure is not merely an abstract isomorphism: the flip and lift-coassociativity axioms force it to be the diagonal, which exists only in rank ≤ 1 (Theorem 13). So the infinite abstractly solid part $\mathbb{N} \cdot y$ is a *phantom* — abstractly solid but not a structure — and uniqueness holds at every rank.

Assembling the three with the forward correspondence (both surviving parts are realised — id and ∂) gives Theorem 1.

Finiteness is not load-bearing: abstract solidity versus a tangent structure. The rank count above — the step $1+2n = 1+n+n^2 \Rightarrow n \in \{0, 1\}$ — genuinely uses finiteness: the equation $1 + 2\kappa = 1 + \kappa + \kappa^2$ holds for every infinite cardinal κ, so it does not by itself exclude $M = \mathbb{N} \cdot y$. One might therefore fear, as an earlier draft did, that finiteness is sharp and $\mathbb{N} \cdot y$ a third solid structure. It is not, and the fear rested on measuring solidity by an abstract isomorphism. It is true that $\mathbb{N} \cdot y$ is *abstractly solid*: the Cantor pairing $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$ supplies an isomorphism $M \otimes M \cong M$ — but this is an *abstract* isomorphism, not a vertical lift ℓ_M. A tangent structure demands more. Its vertical lift must be *flip-compatible* ($c\ell = \ell$) and *coassociative*, and these two axioms force ℓ_M to be the diagonal $M \to M \otimes M$ — which is surjective only when $\operatorname{rank} M \leq 1$. Hence $\operatorname{rank} M \leq 1$ *at every cardinality*: $\mathbb{N} \cdot y$ carries no tangent structure, and uniqueness holds for a free infinitesimal part of any rank (§5, where this is proved in full). Abstract solidity is thus strictly weaker than underlying a tangent structure once the rank is infinite, and the two coincide (landing in $\{0, 1\}$) at finite rank. The honest statement is the rank-free one; finiteness was guarding a phantom.

Certification. The arithmetic of both halves is machine-checked in a Mathlib-free Lean 4 development with no classical axioms: the idempotent collapse $n^2 = n \iff n \in \{0, 1\}$, the finite enumeration of solid ranks, the three coproduct-rank identities underwriting the "ranks add, never subtract" claim, and the universality collapse $1 + 2n = 1 + n + n^2 \iff n \in \{0, 1\}$ by additive cancellation. The Lean certifies the *arithmetic*; the structural inputs (the universality axiom, that $\lhd$ preserves pullbacks, that M is linear) are cited framework facts, and we are explicit about the boundary (§4, §6).

Contribution. The contributions are: the uniqueness theorem itself (Theorem 1), proved over a rig with neither a PID hypothesis nor subtraction; the rigidity half (finite solid parts $= \{0, y\}$); the rig-internal converse, whose single new idea is that the vertical bundle is a direct sum of free summands so that ranks add, never subtract (§6); the technical observation that on the linear layer the base-change tensor $\otimes W$, the substitution tensor $\lhd$ and the Dirichlet tensor all coincide (Lemma 7), which is simultaneously what makes solidity computable in the bundle's own tensor and what connects the Weil base-change classification to the container derivative; the observation that solidity is strictly weaker than underlying a tangent structure at infinite rank, so that uniqueness is

rank-free (§5); and the Lean certification of the finite arithmetic. The result is a clean deliverable for the tangent-category and `Poly` communities: *on the linear layer of finite-support polynomial functors — the finite free $\mathbb{N}$-modules — the container derivative is the only nontrivial representable first-order tangent structure.*

Outline. §2 recalls containers, the substitution monoidal structure, infinitesimal objects, the representable tangent bundle and the universality of the vertical lift. §3 proves the three-tensor coincidence on the linear layer. §4 proves the rank rigidity (finite solid parts $= \{0, y\}$) and records its Lean certification. §5 shows that the infinite-rank solid part $\mathbb{N} \cdot y$ is not a tangent structure, so uniqueness is rank-free. §6 proves the solidity converse from vertical-lift universality — the main new content — and records its Lean certification. §7 assembles the uniqueness theorem. §8 compares with Lanfranchi–Lemay. §9 concludes with the precedent and the sequel.

2 Preliminaries

We recall only what the argument uses; for the Cartesian differential and tangent structure on SPoly we refer to the companion note [11].

Containers and the substitution monoidal category. A *container in n variables* is a set S of *shapes* together with, for each $s \in S$ and each $i \in \{1, \dots, n\}$, a position set P_s^i; its extension is the polynomial functor $[\![S \lhd P]\!](\vec{X}) = \sum_{s \in S} \prod_{i=1}^{n} X_i^{P_s^i}$. We work with finitary containers, so only the cardinalities $|P_s^i|$ matter up to isomorphism [1]. Composition of containers is substitution of extensions, written $G \lhd F$, with unit the identity container y; this makes $(\mathsf{Poly}, \lhd, y)$ a (strict, once we pass to isomorphism classes) monoidal category, and the comonoids in it are exactly the small categories [2, 8]. A container is *linear* if it is of the form $S \cdot y$ for a set S, i.e. its extension is $\vec{X} \mapsto S \times X$ in one variable; *finite-support* if its shape set is finite. We write SPoly for the finite-support polynomial functors.

The linear layer $\mathsf{Lin}(\mathsf{SPoly})$ and the free $\mathbb{N}$-modules $\mathsf{FreeMod}_{\mathbb{N}}$. The classification lives not in SPoly at large but in its subcategory of *linear* maps. Let $\mathsf{Lin}(\mathsf{SPoly})$ be the (wide) subcategory with the same objects $n \in \mathbb{N}$ but only the *linear* morphisms $n \to m$: those given by an $m \times n$ matrix of natural numbers (equivalently, tuples of finite-support linear containers $S \cdot y$), with composition the matrix product. Write $\mathsf{FreeMod}_{\mathbb{N}}$ for the category of *all* free $\mathbb{N}$-modules and $\mathbb{N}$-linear maps; the functor $n \mapsto \mathbb{N}^n$ is an equivalence of $\mathsf{Lin}(\mathsf{SPoly})$ onto its full subcategory of *finite-rank* modules. Finite-support SPoly sees only finite ranks; the uniqueness theorem is proved in the larger $\mathsf{FreeMod}_{\mathbb{N}}$, so that "no exotic structure at infinite rank" is a theorem and not a blind spot of the finiteness restriction (§5). What matters for the morphisms is this: the augmentation $\varepsilon \mapsto 0$ of a Weil algebra and the other structure maps below are $\mathbb{N}$-matrices — legitimate morphisms of $\mathsf{FreeMod}_{\mathbb{N}}$ — even though, read naively as container maps in $(\mathsf{Poly}, \lhd, y)$, $\varepsilon \mapsto 0$ is not a container morphism. On $\mathsf{Lin}(\mathsf{SPoly})$ the Cartesian differential / tangent structure of SPoly [11] restricts, and there it is computed by base change (Lemma 7), so a linear container $S \cdot y$ appears as a *morphism* (1×1 matrix $[\,|S|\,]$) and $\lhd$ appears as *composition*, not as a tensor of objects.

Infinitesimal objects and representable tangent bundles. We use the representable presentation of tangent structure of [9], transported along the dictionary $(\mathsf{Mod}_R, \otimes_R, R) \rightsquigarrow (\mathsf{Mod}_{\mathbb{N}}, \otimes_{\mathbb{N}}, \mathbb{N})$: the *same* module/algebra layer as Lanfranchi–Lemay, but over the rig $\mathbb{N}$ in place of a commutative ring. The tangent functor is *base change by a Weil algebra*, not substitution; the link to Poly is that

the finite-rank free $\mathbb{N}$-modules are exactly the objects of $\mathsf{Lin}(\mathsf{SPoly})$, on which base change coincides with $\lhd$ (Lemma 7).

Definition 2 (Infinitesimal object / Weil algebra)**.** An *infinitesimal object* is an augmented square-zero $\mathbb{N}$-algebra $D = y \oplus M$: a pointed object with unit/zero $z\colon y \to D$ and augmentation $p\colon D \to y$, $p \circ z = \mathrm{id}_y$, *infinitesimal part* $M = \ker(p)$ a free $\mathbb{N}$-module with $M \cdot M = 0$ inside D (the $\varepsilon^2 = 0$ condition). The associated tangent functor is *base change* $T := (-) \otimes_{\mathbb{N}} D$, with tangent projection $A \otimes p\colon A \otimes D \to A$ and zero section $A \otimes z\colon A \to A \otimes D$. The dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2$ are the first such object. We do *not* rank M finite here: the classification below holds at every rank.

Remark 3 (Base change is not the substitution formula)**.** It is worth stating once, cleanly, what base change $T = (-) \otimes W$ is and is not. On the free module $\mathbb{N}^n$ it is $\mathbb{N}^n \mapsto \mathbb{N}^n \otimes_{\mathbb{N}} W = \mathbb{N}^n \oplus \varepsilon\mathbb{N}^n$, doubling the rank; the ring map $\mathbb{N}[y] \to W[y]$ underlying it sends $y \mapsto y$ and $y^2 \mapsto y^2$. This is distinct from the dual-number *substitution* $y \mapsto y + \varepsilon v$, which sends $y^2 \mapsto y^2 + 2\varepsilon vy$: the substitution is the unit of the adjunction between $(-) \otimes W$ and the Kähler functor, and it is how the companion note [11] computes the combinator ∂ on SPoly *maps* (where a polynomial is a morphism). The two agree on *linear* arguments — which is all the tangent *functor* on $\mathsf{FreeMod}_{\mathbb{N}}$ ever sees — and a non-linear polynomial is neither an object nor a morphism of $\mathsf{FreeMod}_{\mathbb{N}}$, so no confusion arises inside the module category.

Definition 4 (Representable first-order tangent structure)**.** A *representable first-order tangent structure* is a tangent structure (in the sense of Cockett–Cruttwell [7], first order) whose bundle functor is base change $T = (-) \otimes D$ by an infinitesimal object $D = y \oplus M$. Its infinitesimal part M is called *solid* if the *vertical-lift component* $\ell_M\colon M \to M \otimes M$ is an isomorphism.

Remark 5 (Why solidity is ℓ_M, not μ)**.** The square-zero condition $M \cdot M = 0$ makes the multiplication $\mu\colon M \otimes M \to M$ the *zero* map (e.g. $\mu(\varepsilon \otimes \varepsilon) = \varepsilon^2 = 0$), so μ is never an isomorphism except in the degenerate case $M = 0$; an earlier draft's "μ an isomorphism" was therefore the wrong map. The map that is forced to be an isomorphism is the vertical-lift component $\ell_M\colon M \to M \otimes M$ (for W, $\varepsilon \mapsto \varepsilon_1\varepsilon_2$); universality of the vertical lift forces exactly this (§6). Only the abstract isomorphism $M \otimes M \cong M$ it provides is used below, so the rigidity computation is unchanged.

Remark 6 (Convention: $\otimes$ versus $\lhd$, and why it matters)**.** The tangent functor is base change $(-) \otimes D$, never substitution $(-) \lhd D$ on all of Poly. The distinction is real: full $\lhd$ has no $\varepsilon^2 = 0$ truncation (for $D = W$, $y^2 \lhd W = y^2 \lhd 2y = (2y)^2 = 4y^2 \neq 3y^2$, and $y^n \lhd W = W^n$ is multiplicative), so $(-) \lhd W$ is *not* the dual-number substitution and, as one checks, forces the trivial structure alone. What saves the container reading is Lemma 7: on the *linear layer* $S \cdot y$ — the only layer the rank arithmetic of §4–§6 ever touches — base change $\otimes$, substitution $\lhd$ and the Dirichlet tensor all agree. Every infinitesimal object below ($D = y \oplus M$, the second-order object D_2, the vertical object P') is a free module, so Lemma 7 lets us compute it by whichever of the three is convenient. We do these computations inside $\mathsf{FreeMod}_{\mathbb{N}}$, where $\lhd$ is matrix composition and the structure maps (augmentation, zero, $\hat{g}$) are $\mathbb{N}$-matrices — which is why the "quick substitution formula" is legitimate: the maps in sight are genuine morphisms of $\mathsf{FreeMod}_{\mathbb{N}}$, not container maps of $(\mathsf{Poly}, \lhd, y)$. The tangent *functor* itself remains base change $(-) \otimes D$.

The dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2 = y \oplus \langle \varepsilon v \rangle$ are such an object: the augmentation is $\varepsilon \mapsto 0$ ($p\colon W \to y$) and the zero is $y \hookrightarrow W$ (z). Base change $(-) \otimes W$ on $\mathsf{FreeMod}_{\mathbb{N}}$ is the module-level tangent functor; its restriction is the ∂-structure of [11] on the linear layer (Remark 3). Its infinitesimal part is the rank-1 free module $M = \langle \varepsilon v \rangle$; the trivial structure is $D = y$, $M = 0$, $T = \mathrm{id}$. That base change by a Weil algebra is a genuine tangent structure — not merely an endofunctor

— is standard (Cockett–Cruttwell [7]; the general representable case, [9]); that its combinator on SPoly is the container derivative ∂ is [11]. We take both as established and ask only whether other infinitesimal objects give rise to further structures.

The universality of the vertical lift. Among the tangent-structure axioms of [7] we use one by name. The *second tangent bundle* is $T_2 := T \times_{\mathrm{id}} T$ (the pullback of $T \xrightarrow{p}$ id along itself), and the *vertical bundle* V is the pullback of the tangent map $Tp\colon T^2 \to T$ along the zero $0\colon \mathrm{id} \to T$. The *vertical lift* is a canonical comparison $T_2 \to T^2$, and the axiom of *universality of the vertical lift* ([7], part of the definition of a tangent structure) asserts that the square

$$\begin{array}{ccc} T_2 & \longrightarrow & T^2 \\ \downarrow & & \downarrow Tp \\ \mathrm{id} & \xrightarrow{0} & T \end{array}$$

is a pullback; equivalently, that the induced map $T_2 \to V$ is an isomorphism. This is the single structural axiom the converse of §6 invokes.

The infinitesimal part is a free module. Inside $\mathsf{FreeMod}_{\mathbb{N}}$ there is nothing to force: the objects *are* the finite free $\mathbb{N}$-modules, so the infinitesimal part $M = \ker(p)$ of any infinitesimal object is a free $\mathbb{N}$-module, with a well-defined rank (Lemma 9). This is what supplies freeness in §4, and it replaces — indeed trivialises — Lanfranchi–Lemay's projective-to-free step, which over a ring needs a PID.

3 Three tensors coincide on the linear layer

This is the linchpin of the whole note, and it carries two burdens at once. First, it reconciles the two readings of the tangent functor: the *bundle* is base change $\otimes W$, whereas the container derivative is phrased with substitution $\triangleleft$, and these are genuinely different operations on Poly (Remark 6). Second, the *solid* condition $M \otimes M \cong M$ lives in the "module layer" where rank multiplies, which looks like yet a third product. The following observation collapses all three onto the linear layer.

Lemma 7 (Three tensors coincide on the linear layer)**.** *For linear modules, base change by a linear object, substitution, and the Dirichlet product all agree:*

$$(S \cdot y) \otimes (T \cdot y) \;=\; (S \cdot y) \triangleleft (T \cdot y) \;=\; (S \times T) \cdot y \;=\; (S \cdot y) \otimes_{\mathrm{Dir}} (T \cdot y).$$

Proof. The extension of $S \cdot y$ is $X \mapsto S \times X$. Composing, $(S \cdot y) \triangleleft (T \cdot y)\colon X \mapsto S \times (T \times X) = (S \times T) \times X = (S \times T) \cdot y$. The Dirichlet product of $\sum_s y^1$ and $\sum_t y^1$ is $\sum_{(s,t)} y^{1\times 1} = (S \times T) \cdot y$. Base change of the free module $S \cdot y$ by the free module $T \cdot y$ tensors the generating sets, $(S \times T) \cdot y$. All three equal $(S \times T) \cdot y$. □

Remark 8. The identity is special to the linear layer; off it the three products diverge. Substitution and Dirichlet already part on two-term polynomials ($(y{+}1) \triangleleft (y{+}1) = y{+}2$ while $(y{+}1) \otimes_{\mathrm{Dir}} (y{+}1) = y + 3$), and base change parts from substitution precisely at the $\varepsilon^2 = 0$ truncation: $y^2 \triangleleft (2y) = 4y^2$ whereas $y^2 \otimes W$ reads off $y^2 + 2\varepsilon v y$. So Lemma 7 is genuinely a statement about the linear layer, not a global identity — which is exactly why the uniqueness theorem is a statement about that layer. None of the three is the "doubling" Dirichlet action $(2y) \otimes_{\mathrm{Dir}} (-)$, which sends $p \mapsto 2p$ and has nothing to do with the derivative.

Consequence. On the linear layer the bundle's own base-change tensor $\otimes$ already multiplies rank, and agrees with $\triangleleft$, so the solid condition $M \otimes M \cong M$ reads $|S|^2 = |S|$ in the very tensor the bundle uses — and the entire rank arithmetic of §4–§6, taking place on this layer, may be read interchangeably in any of the three. This single coincidence is what makes the port faithful and what licenses the container-derivative reading of a result about Weil base change.

4 Rank rigidity: the solid parts are $\{0, y\}$

We classify the finite solid infinitesimal parts. Two structural facts about linear containers, then the rank computation.

Lemma 9 (Freeness is automatic over $\mathbb{N}$)**.** *Every linear container $S \cdot y$ is the free $\mathbb{N}$-module on the set S; two linear containers are isomorphic iff their shape sets are equinumerous. The cardinal $|S| =: \mathrm{rank}(M)$ is a complete isomorphism invariant.*

Proof. Under coproduct of containers and the $\mathbb{N}$-action $n \cdot (S \cdot y) = (n \times S) \cdot y$, the linear containers form the free-$\mathbb{N}$-module functor applied to sets: $S \cdot y$ is the $|S|$-fold copower of y. A container isomorphism $S \cdot y \cong T \cdot y$ is, since morphisms between linear containers are just functions on shapes (positions being singletons), a bijection $S \cong T$. Hence rank is a complete invariant. □

This is the analogue of Lanfranchi–Lemay's "projective ⇒ free ⇒ rank is a complete invariant" (their Lemma 4.16, via Rotman's structure theorem, which *requires* the base to be a PID). Here it is *automatic*: the linear polynomial functors *are* the free $\mathbb{N}$-modules by definition, so no PID hypothesis is needed and the rigidity below holds over the rig $\mathbb{N}$ — strictly outside the Lanfranchi–Lemay hypotheses.

Lemma 10 (Solid ⇒ idempotent rank)**.** *If $M = S \cdot y$ is solid, i.e. the vertical-lift component $\ell_M \colon M \to M \otimes M$ is an isomorphism, then $|S|^2 = |S|$.*

Proof. By Lemma 7, $M \otimes M = M \triangleleft M = (S \times S) \cdot y$, of rank $|S|^2$ by Lemma 9. An isomorphism $\ell_M \colon S \cdot y \cong (S \times S) \cdot y$ forces, by Lemma 9, a bijection $S \cong S \times S$, i.e. $|S| = |S|^2$. □

Theorem 11 (Finite rigidity)**.** *A finite-support solid infinitesimal part has rank in $\{0, 1\}$. Hence there are exactly two: $M = 0$ (rank 0) and $M = y$ (rank 1). By §2 these are the trivial structure and the AAGM dual-number structure, with combinator ∂.*

Proof. For a natural number $n = |S|$ we have $n^2 = n \iff n(n-1) = 0 \iff n \in \{0, 1\}$, since $\mathbb{N}$ has no zero divisors. By Lemma 10 a finite solid part has $|S|^2 = |S|$, so $|S| \in \{0, 1\}$; by Lemma 9 the rank determines M, giving $M \in \{0, y\}$. □

This is the Lanfranchi–Lemay rank computation $R^{n^2} \cong R^n \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$, ported verbatim with the cardinality rank, and — per Lemma 9 — without their PID hypothesis.

Machine certification of the arithmetic. The finite arithmetic of Theorem 11 is certified in a Mathlib-free Lean 4 development with no classical axioms. The oracle proves: $n^2 = n \iff n \in \{0, 1\}$ on $\mathbb{N}$ (in general, `omega`-free); that the solid ranks in $0 \le n \le 12$ are exactly $\{0, 1\}$ (by `decide`); that these two ranks map through the structure assignment to exactly $\{\mathrm{id}, \partial\}$, with the dual-number object of arity 2; that every rank $r \ge 2$ (the Whitney sums of §6) is non-solid; and that the Cantor pairing is injective on a 12×12 window, witnessing $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ for §5. `#print axioms` on the packaged oracle reports only `propext` (a structural, non-classical axiom); the finite `decide` claims are individually axiom-free, with no `sorryAx`, `Quot.sound` or `Classical.choice`.

5 Infinite rank: solidity without a tangent structure

The rank count of Theorem 11 — and of the converse's step $1+2n = 1+n+n^2 \Rightarrow n \in \{0,1\}$ (§6) — used finiteness, because $\kappa^2 = \kappa$ for every infinite cardinal κ (under choice). One might therefore expect the finite-rank hypothesis to be sharp, with $\mathbb{N} \cdot y$ a third solid structure. It is not. The expectation conflates two distinct notions of solidity, which we name once and keep apart. Call M *abstractly solid* if there merely *exists* an $\mathbb{N}$-module isomorphism $M \cong M \otimes M$ (equivalently the cardinality condition $|S|^2 = |S|$), and *vertical-lift solid* if the structural vertical-lift component $\ell_M \colon M \to M \otimes M$ of a tangent structure — the map of Lemma 10 — is an isomorphism. At finite rank the two coincide, both landing in $\{0,1\}$. At infinite rank they part: an infinite free module *is* abstractly solid but, as we now show, is never vertical-lift solid, because the tangent axioms force ℓ_M to be the diagonal. So $\mathbb{N} \cdot y$ passes the abstract test and fails the one that matters. Uniqueness is rank-free.

Proposition 12 ($\mathbb{N} \cdot y$ is abstractly solid)**.** *The countable free module $M = \mathbb{N} \cdot y$ is* abstractly *solid: the Cantor pairing supplies a bijection $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$, hence an $\mathbb{N}$-module isomorphism $M \cong (\mathbb{N} \times \mathbb{N}) \cdot y = M \otimes M$ (Lemma 7), so $M \otimes M \cong M$ with $\operatorname{rank} M = |\mathbb{N}| \notin \{0,1\}$. This isomorphism is* abstract*: it is some module iso, not the vertical-lift component of a tangent structure. Abstract solidity therefore does not by itself bound the rank — and, as Theorem 13 shows, it does not supply a vertical lift either.*

So the rank rigidity (Theorem 11) is about *finite* solid parts, and its hypothesis is genuinely used. The point of this section is that this does *not* leave a gap in the uniqueness theorem, because the extra axioms of a tangent structure exclude $\mathbb{N} \cdot y$ outright.

Theorem 13 (Rank-free exclusion of infinite parts)**.** *Let $T = (-) \otimes D$ be a representable first-order tangent structure with $D = y \oplus M$, M a free $\mathbb{N}$-module of* any *rank. Then $\operatorname{rank} M \leq 1$. In particular $\mathbb{N} \cdot y$ carries no first-order tangent structure, and the only infinitesimal objects underlying one are y and W.*

The one-sentence reason is that the isomorphism ℓ_M of a tangent structure is not merely abstract but *is* the vertical lift, and the flip axiom $c\ell = \ell$ together with lift-coassociativity force ℓ_M to be the diagonal $M \to M \otimes M$ — which, being an isomorphism onto $M \otimes M$, exists only when $\operatorname{rank} M \leq 1$. We give the argument in full.

Proof. Because a natural transformation of base-change functors $(-) \otimes V \Rightarrow (-) \otimes V'$ is, by evaluation at the unit y, the same datum as an $\mathbb{N}$-module map $V \to V'$, the vertical lift $\ell \colon T \Rightarrow T^2$ and the canonical flip $c \colon T^2 \Rightarrow T^2$ of the tangent structure are

$$\ell_A = \mathrm{id}_A \otimes \delta, \qquad c_A = \mathrm{id}_A \otimes \gamma$$

for fixed module maps $\delta \colon D \to D \otimes D$ and $\gamma \colon D \otimes D \to D \otimes D$. Decompose $D \otimes D = y \oplus M_1 \oplus M_2 \oplus (M \otimes M)$ with $M_1 = M \otimes y$, $M_2 = y \otimes M$, and $M \otimes M$ the mixed part of rank $(\operatorname{rank} M)^2$. Fix a basis $\{e_k\}_{k \in K}$ of M, so $\operatorname{rank} M = |K|$. We argue in five steps; none uses finiteness.

(0) The lift lands in the mixed part: $\delta(M) \subseteq M \otimes M$. Two of the tangent-bundle axioms, $p_T \circ \ell = 0 \circ p$ and $T(p) \circ \ell = 0 \circ p$ (with $p \colon D \to y$ the augmentation and 0 the zero section), restrict on M to $(p \otimes \mathrm{id}) \circ \delta = 0$ and $(\mathrm{id} \otimes p) \circ \delta = 0$. Since $\ker(p \otimes \mathrm{id}) \cap \ker(\mathrm{id} \otimes p) = M \otimes M$ inside $D \otimes D$, this gives $\delta(M) \subseteq M \otimes M$. Write $\ell_M := \delta|_M \colon M \to M \otimes M$ for the resulting *vertical-lift component*.

(1) Universality makes ℓ_M an isomorphism. By the universality of the vertical lift (§2) the comparison $v \colon T_2 \to V$ is an isomorphism of functors. Evaluated at y and written in coordinates,

$v(a;u,w) = a + \sum_k u_k e_k + \sum_k w_k\, \ell_M(e_k)$; as a map of fibres $y \oplus M \oplus M \to y \oplus M \oplus (M \otimes M)$ it is the identity on the summand $y \oplus M$ and ℓ_M on the last summand. Hence v is an isomorphism iff $\ell_M \colon M \to M \otimes M$ is: universality *produces* the solidity of Definition 4, rather than assuming it, and it refines the rank shadow $T_2 \cong V$ (i.e. $1 + 2n = 1 + n + n^2$) of §6 to an honest isomorphism of the lift map itself.

(2) Over $\mathbb{N}$ an isomorphism of free modules permutes bases. A free $\mathbb{N}$-module is the free commutative monoid on its basis, whose irreducible elements (those with $x = a + b \Rightarrow a = 0$ or $b = 0$) are exactly the basis vectors — no proper multiple $2e_k = e_k + e_k$ is irreducible. A monoid isomorphism carries irreducibles to irreducibles bijectively, so ℓ_M sends each basis vector to a basis vector of $M \otimes M$:

$$\ell_M(e_k) = e_{\beta_1(k)} \otimes e_{\beta_2(k)}, \qquad \beta := (\beta_1, \beta_2) \colon K \to K \times K \text{ a bijection.}$$

This holds at every cardinality — which is why the abstract escape "$\kappa = \kappa^2$ lets ℓ_M be an isomorphism at $\aleph_0$" yields nothing structured.

(3) Flip-compatibility forces $\gamma = \mathrm{id}$ on the mixed part. The axiom $c \circ \ell = \ell$ gives $\gamma \circ \ell_M = \ell_M$; since ℓ_M is surjective onto $M \otimes M$ (step 1), this forces $\gamma = \mathrm{id}$ on $M \otimes M$ — *whatever* the flip is. So for rank $M \geq 2$ the flip is *not* the swap but is forced to act as the identity on the mixed part; this is the only configuration compatible with an invertible ℓ_M, and there is no exotic non-representable flip to escape through.

(4) Lift-coassociativity makes ℓ_M strictly coassociative. The vertical-lift coassociativity coherence of [7] reads $T(\ell) \circ \ell = (c\text{-twist}) \circ \ell_T \circ \ell$. Both composites land in the fully mixed part $M \otimes M \otimes M$ (each application of δ maps M into $M \otimes M$, step 0). The c-twist is a composite of flips on adjacent tensor factors; on $M \otimes M \otimes M$ each acts through $\gamma|_{M \otimes M} = \mathrm{id}$ (step 3), so the c-twist term is idle — it acts as the identity there — and the coherence collapses to

$$(\ell_M \otimes \mathrm{id}) \circ \ell_M = (\mathrm{id} \otimes \ell_M) \circ \ell_M,$$

strict coassociativity of ℓ_M. The conclusion is robust to the exact form of the twist: whatever composite of flips the axiom places there, each is the identity on the mixed part by step 3.

(5) A coassociative basis-bijection is the diagonal. Substituting $\ell_M(e_k) = e_{\beta_1(k)} \otimes e_{\beta_2(k)}$ into strict coassociativity and comparing basis triples gives

$$\beta_1 \circ \beta_1 = \beta_1, \qquad \beta_2 \circ \beta_2 = \beta_2, \qquad \beta_2 \circ \beta_1 = \beta_1 \circ \beta_2.$$

Since β is a bijection onto $K \times K$ its first coordinates exhaust K, so β_1 is surjective; a surjective idempotent on a set is the identity, whence $\beta_1 = \mathrm{id}$, and likewise $\beta_2 = \mathrm{id}$. Then $\beta(k) = (k, k)$ is the diagonal, surjective onto $K \times K$ only when $|K| \leq 1$. Hence $\operatorname{rank} M = |K| \leq 1$. The surviving cases are $\operatorname{rank} M = 0$ ($M = 0$, $T = \mathrm{id}$) and $\operatorname{rank} M = 1$ ($M = y$, $T = \partial$), where the diagonal is all of $K \times K$.[1] No step used finiteness, so the bound holds at every cardinality. □

Remark 14 (An independent route in the representable case)**.** When the flip is $\lhd$-monoidal — γ an algebra automorphism of $D \otimes D$ — fixing y and exchanging $M_1 \leftrightarrow M_2$ forces $\gamma(e_i \otimes e_j) = e_{\psi(j)} \otimes e_{\varphi(i)}$, a swap-type map that equals the identity on $M \otimes M$ only when $\operatorname{rank} M \leq 1$. Combined with step (3) this already gives $\operatorname{rank} M \leq 1$ *without* invoking lift-coassociativity. So in the representable setting the exclusion closes by two independent routes; lift-coassociativity (steps 4–5) is what additionally rules out the exotic non-monoidal flip.

[1] At finite rank ≥ 2 there is in any case no bijection $K \cong K \times K$ at all — $|K| < |K|^2$ — so the diagonal cannot be surjective and the obstruction is already immediate from cardinality, with nothing to search over. The work of steps (4)–(5) is to make the bound hold *uniformly*, in particular at infinite rank, where $\kappa = \kappa^2$ and cardinality alone decides nothing. The finite-rank arithmetic $1 + 2n = 1 + n + n^2 \Rightarrow n \in \{0, 1\}$ is Lean-certified (§4).

The exclusion is thus a theorem about the vertical-lift *data*, not about the abstract isomorphism type of M.

Remark 15 (Solidity versus structure, and the contrast with Lanfranchi–Lemay)**.** The correspondence ($\star$) of §6 — tangent structure $\iff$ solid part — is therefore a bijection *only* at finite rank, where both sides land in $\{0,1\}$. At infinite rank the forward direction fails: $\mathbb{N}\cdot y$ is abstractly solid but is not a structure (Proposition 12, Theorem 13). This is where our setting diverges from Lanfranchi–Lemay. Over a ring their extra solid parts at non-PID / non-finitely-generated bases (their Examples 4.20–4.21) are genuine, so they must impose finite generation (Niefield coexponentiability) to recover uniqueness. Over $\mathbb{N}$ the extra solid part $\mathbb{N}\cdot y$ is a *phantom*: it is abstractly solid, but the flip and coassociativity axioms refuse it a tangent structure, so uniqueness needs no finiteness hypothesis at all. An earlier draft of this note read $\mathbb{N}\cdot y$ as a genuine third solid structure and declared "finiteness is sharp"; that reading mistook abstract solidity for vertical-lift solidity, and is withdrawn.

6 The solidity converse: universality of the vertical lift

Theorem 11 classifies the solid infinitesimal *parts*. To pass from parts to tangent *structures* we need the converse direction of the correspondence

> ($\star$) representable first-order tangent structure $T = (-)\otimes D \iff$ solid infinitesimal part $M = \ker(D \to y)$.

The forward direction is immediate once the parts are known (§7). The converse — that every representable first-order tangent structure has a *solid* infinitesimal part — is Lanfranchi–Lemay's Theorem 4.8, proved over a commutative ring, and is the step that did not transfer verbatim: their derivation computes a vertical bundle by a short exact sequence, hence by subtraction, which the rig $\mathbb{N}$ lacks. We now give a rig-internal proof. The engine is the universality of the vertical lift (§2); the one new idea is that the vertical bundle is a *direct sum* of its surviving summands, so its rank is computed by addition and never by a difference — rank is additive over direct sums of free $\mathbb{N}$-modules, and $\mathbb{N}$ has no subtraction.

The entire argument lives within the finite-rank free modules $\mathsf{Lin}(\mathsf{SPoly})$, where by Lemma 7 the tangent functor $T = (-)\otimes D$ *coincides* with substitution $(-)\triangleleft D$ — base change of a free module by a free module is substitution. We therefore compute the objects below with the quick substitution formula $\triangleleft$, which preserves pullbacks pointwise (Lemma 16). This is legitimate because all the maps in sight are morphisms of $\mathsf{FreeMod}_{\mathbb{N}}$ — the augmentation $\hat{p}$ and zero $\hat{z}$ are $\mathbb{N}$-matrices, and $\triangleleft$ between free modules is their matrix composition — so the pullbacks and the rank counts are taken in $\mathsf{FreeMod}_{\mathbb{N}}$, not in $(\mathsf{Poly}, \triangleleft, y)$ (where $\hat{p}$, sending $\varepsilon \mapsto 0$, is not a container morphism at all). Here $D = y \oplus M$, $M = S\cdot y$ a free module of finite rank $n = |S|$ (the finiteness used in this section; the infinite case is handled by Theorem 13).

6.1 Substitution preserves pullbacks on both sides

Lemma 16 ($\triangleleft$ preserves pullbacks)**.** *For every $D \in \mathsf{Poly}$ the functor $(-)\triangleleft D$ preserves pullbacks; and for every polynomial A the functor $A \triangleleft (-)$ preserves pullbacks.*

Proof. Pullbacks in Poly are computed pointwise. For $(-)\triangleleft D$: if $F = G \times_H K$ pointwise then for all X, $(F\triangleleft D)(X) = F(D(X)) = G(D(X)) \times_{H(D(X))} K(D(X)) = (G\triangleleft D \times_{H\triangleleft D} K \triangleleft D)(X)$, so $F\triangleleft D = G\triangleleft D \times_{H\triangleleft D} K\triangleleft D$. For $A\triangleleft(-)$: a polynomial functor A preserves connected

limits ([8], Prop. 1.9), and a pullback is a connected limit, so A preserves the pointwise pullback $E_1(X) \times_{E_0(X)} E_2(X)$; hence $A \lhd (-)$ preserves pullbacks. □

Consequence: the tangent data are all right-tensorings. The projection, zero, tangent map and the two bundles of $T = (-) \otimes D$ are $(-) \otimes$ (structure map on D) — which on the linear layer is $(-) \lhd$ (structure map): $p = (-) \lhd \hat{p}$ with $\hat{p}\colon D \to y$ the augmentation and $z = (-) \lhd \hat{z}$ with $\hat{z}\colon y \to D$ the zero; and, by Lemma 16 applied so that $A \lhd (-)$ carries the relevant pointwise pullback,

$$T_2 = (-) \otimes D_2, \quad D_2 := D \times_y D; \qquad V = (-) \otimes P', \quad P' := \hat{g}^{-1}(\operatorname{im} \hat{z}), \quad \hat{g} := \hat{p} \lhd D\colon D \lhd D \to D.$$

All of D, D_2, $D \lhd D$, P' are linear objects, and y is one of them.

6.2 The three coproduct counts

We work in the based setting: ε-grading modulo ε^2 within each copy of D, with the mixed term surviving across copies — the Weil-algebra model $\mathbb{N}[\varepsilon_1, \dots]/$(within-copy squares) that the base-change presentation of T [11] delivers. This is the content of "$T = (-) \otimes D$ is base change," and it is what makes D_2 (fibre product, no mixed term) differ from the composite $D \lhd D$ (mixed term present): the $\varepsilon^2 = 0$ truncation lives in D, not in the composition. Each object below is a direct sum of free summands, so its rank is the *sum* of the summand ranks (rank is additive over $\oplus$, and $\mathbb{N}$ has no subtraction to undo the sum).

Lemma 17 (Coproduct ranks)**.** *With $n = \operatorname{rank} M$ and $\operatorname{rank}(M \lhd M) = n^2$ (Lemmas 7, 9):*

$$\begin{aligned} D \lhd D &= y \oplus M \oplus M \oplus (M \lhd M), && \text{rank} = 1 + n + n + n^2 = (1+n)^2; \\ D_2 = D \times_y D &= y \oplus M \oplus M, && \text{rank} = 1 + 2n; \\ P' &= y \oplus \varepsilon_{\text{out}} \oplus (M \lhd M), && \text{rank} = 1 + n + n^2. \end{aligned}$$

Proof. *($D \lhd D$.)* Since M is linear, $M(X) = S \times X$, so $(D \lhd D)(X) = D(D(X)) = X \oplus M(X) \oplus M(X) \oplus M(M(X))$: the base y, the inner copy's M (ε_{in}), the outer copy's M (ε_{out}), and the mixed $M \lhd M = (S \times S) \cdot y$. Ranks $1, n, n, n^2$.

(D_2.) A fibre product over the base shares y and adjoins the two infinitesimal directions but creates *no* product of the two infinitesimals: $D_2 = y \oplus M \oplus M$, rank $1 + 2n$.

(P'.) The map $\hat{g} = \hat{p} \lhd D$ applies the augmentation $\hat{p}$ ($\varepsilon \mapsto 0$) to the *outer* copy: it kills ε_{out} and the mixed term $M \lhd M$ (both carry an outer-M factor) and fixes y and ε_{in}. The zero $\hat{z}$ has image the y-summand. Hence $P' = \hat{g}^{-1}(\operatorname{im} \hat{z})$ is the sub-coproduct on which the ε_{in}-component vanishes: $P' = y \oplus \varepsilon_{\text{out}} \oplus (M \lhd M)$, rank $1 + n + n^2$. Because $\hat{g}$ and $\hat{z}$ respect the direct-sum decomposition — $\hat{g}$ acts summand-wise and $\operatorname{im} \hat{z}$ is the y-summand — this pullback *is* the direct sum of the surviving summands, so its rank is the sum $1 + n + n^2$ — not $\operatorname{rank}(D \lhd D) - \operatorname{rank}(\operatorname{im} \hat{z})$. This is the step where Lanfranchi–Lemay form a difference of ranks (a short exact sequence over a ring), and where we do not. □

Both $D_2 = (1 + 2n) \cdot y$ and $P' = (1 + n + n^2) \cdot y$ are linear (a coproduct $y \oplus$ (free) is again linear), so Lemma 9 applies to them.

6.3 Universality forces $n^2 = n$

Theorem 18 (Solidity converse)**.** *Let $T = (-) \otimes D$ be a representable first-order tangent structure on finite-rank $\mathbb{N}$-modules, with $D = y \oplus M$ and $n = \operatorname{rank} M$ finite. Then $n^2 = n$, hence $n \in \{0, 1\}$; i.e. M is solid and $M \in \{0, y\}$.*

Proof. By the universality of the vertical lift (§2) — the same input as step (1) of Theorem 13 — the comparison $T_2 \to V$ is an isomorphism of endofunctors, naturally in the argument. By Lemma 16 and its consequence, $T_2 = (-) \otimes D_2$ and $V = (-) \otimes P'$. Evaluate the functor isomorphism at the unit object y and use $y \otimes E = E$:

$$D_2 \;\cong\; P'.$$

Both are linear (Lemma 17), so by Lemma 9 their shape-counts agree:

$$1 + 2n \;=\; 1 + n + n^2.$$

Now use only that $\mathbb{N}$ is a cancellative rig with no zero divisors — *no subtraction, no additive inverses.* Additive left-cancellation of the common summand $1 + n$ gives $n = n^2$. For $n \geq 1$, multiplicative cancellation of n in $n \cdot n = n \cdot 1$ gives $n = 1$; and $n = 0$ satisfies $0^2 = 0$. Hence $n \in \{0, 1\}$, so $M \in \{0, y\}$ by Lemma 9. In each surviving case the vertical-lift component $\ell_M \colon M \to M \otimes M$ is an isomorphism, as one exhibits directly. For $M = 0$ the mixed part $M \otimes M = 0$ and $\ell_M = 0 \colon 0 \to 0$ is the identity. For $M = y$, take the single basis vector ε of $M = \langle \varepsilon v \rangle$; then $M \otimes M$ is free of rank 1 on $\varepsilon \otimes \varepsilon$ (Lemmas 7, 9), and the vertical lift is $\ell_M \colon \varepsilon \mapsto \varepsilon \otimes \varepsilon$ (the $\varepsilon \mapsto \varepsilon_1 \varepsilon_2$ of $W \otimes W$), a bijection of the two one-element bases, hence an isomorphism. So both surviving parts are genuinely solid, as claimed. □

Remark 19 (Why this is rig-internal)**.** The subtraction that blocked a literal port lived in computing the vertical-bundle fibre P' by exactness. In Poly that fibre is a direct sum of the surviving summands (Lemma 17), so its rank is a *sum*. Everything downstream is additive/multiplicative cancellation in $\mathbb{N}$. The derivation never forms a difference of ranks, never inverts an element, and never constructs the solidity isomorphism explicitly — solidity is *output*, read off from $n \in \{0, 1\}$. Lanfranchi–Lemay's Theorem 4.8 instead builds it and argues over a ring; the content we have replaced is precisely their one subtraction.

Remark 20 (The non-solid finite candidates are auxiliary bundles)**.** The finite candidate objects excluded by Theorem 18 are the higher square-zero objects $D_r = y + \varepsilon_1 v_1 + \cdots + \varepsilon_r v_r$ ($\varepsilon_i \varepsilon_j = 0$, $r \geq 2$), with $M = \mathbb{N}^r$, $M \lhd M = \mathbb{N}^{r^2}$, rank $r^2 \neq r$: not solid. These D_r are precisely the Whitney sums $T_r = T \times_X \cdots \times_X T$, which occur in the tangent axioms as *auxiliary* bundles, not as standalone tangent structures. So the objects the converse rejects are exactly the ones that are not primitive structures — solidity coincides with indecomposability, as it should. (The non-solidity of every $r \geq 2$ is part of the certified Lean oracle below.)

Remark 21 (The universality shadow alone needs finiteness)**.** The rank equation $1+2n = 1+n+n^2$ is satisfied by every infinite cardinal κ ($1+2\kappa = \kappa = 1+\kappa+\kappa^2$), so this *rank shadow* of universality does not by itself exclude $M = \mathbb{N} \cdot y$ — which is exactly why Theorem 18 is stated for finite rank. The full vertical-lift data impose more than the rank shadow (the flip and coassociativity axioms), and *that* excludes $\mathbb{N} \cdot y$ at every cardinality (Theorem 13). So the classification is rank-free even though this cleaner universality argument, which only sees ranks, is not; the two agree on the finite range, where both give $\{0, 1\}$.

Machine certification of the arithmetic. The rig-internal arithmetic of Theorem 18 is certified in the same Mathlib-free, axiom-light Lean 4 style. The oracle proves the three coproduct-rank identities of Lemma 17 — $\operatorname{rank}(D \lhd D) = (1+n)^2$, $\operatorname{rank}(D \lhd D) = \operatorname{rank} P' + n$, $\operatorname{rank}(D \lhd D) = \operatorname{rank} D_2 + n^2$ — which are the honesty certificate for "ranks add, never subtract"; the universality collapse $\operatorname{rank} D_2 = \operatorname{rank} P' \iff n \in \{0, 1\}$, proved by `Nat.add_left_cancel` (literal additive cancellation, *no* `omega`) reducing to $n^2 = n \iff n \in \{0, 1\}$; the finite enumeration (survivors

in $0 \le n \le 12$ are exactly $\{0, 1\}$); and the two witnesses $n = 1$ (universality holds, $\operatorname{rank} D_2 = \operatorname{rank} P' = 3$, the Weil rank of $\mathbb{N}[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2, \varepsilon_1\varepsilon_2)$) and $n = 2$ (excluded, $5 \neq 7$). `#print axioms` on the packaged oracle reports only `propext`. The Lean certifies the *arithmetic collapse* and the additivity of the counts; it does *not* formalise the structural inputs — the universality axiom ([7]), that $\triangleleft$ preserves pullbacks (Lemma 16, via [8]), or that the infinitesimal part is a free module (automatic in $\mathsf{FreeMod}_{\mathbb{N}}$) — which are cited framework facts, and we are explicit that they are the mathematical, not the decidable, content.

7 The uniqueness theorem

With both halves in hand the correspondence ($\star$) is an equivalence, and the uniqueness theorem assembles.

Forward: solid ⇒ structure. A finite solid part yields a tangent structure in the two surviving cases. $M = 0$ gives $T = \mathrm{id}$; $M = y$ gives the AAGM dual-number structure, which *is* a tangent structure — the full Cartesian differential / tangent verification is the content of [11]. Since Theorem 11 leaves no third finite solid part, the forward direction never has to manufacture a third structure.

Theorem 22 (Uniqueness)**.** *Among representable first-order tangent structures $T = (-) \otimes D$ on $\mathsf{FreeMod}_{\mathbb{N}}$ (base change by an augmented square-zero $\mathbb{N}$-algebra $D = y \oplus M$, with M a free $\mathbb{N}$-module of* any *rank) there are exactly two: the trivial structure ($D = y$, $T = \mathrm{id}$) and the AAGM dual-number structure ($D = W = \mathbb{N}[\varepsilon]/\varepsilon^2$, combinator ∂). Restricted to the finite-rank part $\mathsf{Lin}(\mathsf{SPoly})$ — the linear layer of SPoly, where $\otimes D$ and $\triangleleft D$ coincide (Lemma 7) — this says the container derivative is the only nontrivial such structure.*

Proof. Let $T = (-) \otimes D$ be such a structure, $D = y \oplus M$. By Theorem 13, $\operatorname{rank} M \le 1$; by Lemma 9 the rank determines M, so $M \in \{0, y\}$ and $D \in \{y,\ W\}$ up to isomorphism. (When M is finite to begin with, the same conclusion follows from the note's own rig-internal converse, Theorem 18, without appeal to the flip argument of §5.) By the forward direction both are realised, by id and ∂ respectively. Hence exactly two. □

Remark 23 (Uniqueness of the full structure, not just the object)**.** Theorem 22 counts infinitesimal *objects*, but the classification is in fact of the full first-order tangent structure $(T, p, 0, +, \ell, c)$. Over $\mathbb{N}$ the remaining data are forced once the object is fixed. For $D = y$ only $T = \mathrm{id}$ is possible. For $D = W$ the additive bundle $+$ is the unique commutative-monoid structure on the rank-1 fibre, the projection and zero are the augmentation and unit of W, and the vertical lift and flip are pinned by $\varepsilon \mapsto \varepsilon_1\varepsilon_2$ and $c\ell = \ell$ (the same forcing, run at rank 1, that excludes higher ranks in Theorem 13). So each of the two infinitesimal objects carries a *unique* first-order tangent structure, and "exactly two" is a statement about structures, not merely objects.

This is Theorem 1, now unconditional and rank-free: the conditional hypothesis of an earlier draft — that every representable first-order tangent structure has a solid infinitesimal part — is discharged by Theorem 18, and the finite-rank hypothesis is removed by Theorem 13.

8 Comparison with Lanfranchi–Lemay

The theorem is a clean deliverable for two audiences. For the tangent-category community [7, 9] it exhibits a rig-based model in which the representable-tangent rigidity *and* its vertical-lift converse

Lanfranchi–Lemay (affine schemes, Mod_R)	This note (Weil base change over $\mathbb{N}$)
tangent functor $T = D \otimes_R (-)$, base change	tangent functor $T = (-) \otimes_{\mathbb{N}} D$, base change ($= \triangleleft$ on the linear layer, Lemma 7)
base a commutative ring R; rigidity needs R a **PID**	base the rig $\mathbb{N}$ (no subtraction); rigidity needs **no** PID
projective $\Rightarrow$ free = Rotman structure theorem (uses PID)	linear module *is* free $\mathbb{N}$-module (Lemma 9) — automatic
rank in R-modules, $R^{n^2} \cong R^n$	cardinality rank, $(S \times S) \cong S$ (Lemmas 7, 9)
rigidity: solid $\ell_M \colon M \cong M \otimes_R M \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$	rigidity: solid $\ell_M \colon M \cong M \otimes M \Rightarrow \lvert S\rvert = \lvert S\rvert^2 \Rightarrow \lvert S\rvert \in \{0, 1\}$
converse: vertical bundle fibre via **short exact sequence** (rank by **subtraction**)	converse: fibre P' a **direct sum** $y \oplus \varepsilon_{\text{out}} \oplus (M \otimes M)$ (rank by **addition**; Lemma 17)
two structures: trivial + Kähler/dual-number	two structures: id + ∂ (Theorem 22)
extra solid parts at non-PID / non-f.g. are **genuine** $\Rightarrow$ finite generation *required* (Ex. 4.20–4.21)	extra solid part $\mathbb{N}\cdot y$ at infinite rank is a **phantom** — solid but no structure (Thm 13), so uniqueness is **rank-free**

Table 1: The port is faithful (same base-change tangent functor, same rank-idempotent mechanism, same vertical-lift converse) and sharper on two axes: freeness is automatic over $\mathbb{N}$ (rigidity over a rig, not a PID), and the vertical bundle is a coproduct (converse by addition, not subtraction). The three-tensor coincidence (Lemma 7), which identifies base change with container substitution on the linear layer, and the additivity of rank over direct sums of free $\mathbb{N}$-modules (Lemma 17) are the linchpins specific to this setting.

hold for strictly weaker reasons than over a PID — no principal ideals, no subtraction — isolating exactly which uses of ring structure in [9] were essential (none, for finite-dimensional infinitesimal parts) and which were incidental (the PID and the short exact sequence, both replaced). For the `Poly` and containers community [1, 2] it upgrades "the container derivative is *a* tangent structure" [11] to "it is the *only* nontrivial one among representable Weil-base-change tangent structures," a uniqueness theorem for a construction that is otherwise defined by a formula.

9 Conclusion: precedent and sequel

We have proved that among the representable first-order tangent structures given by Weil base change $T = (-) \otimes D$ on $\mathsf{FreeMod}_{\mathbb{N}}$ — for a free infinitesimal part of any rank — there are exactly two, the trivial one and the container derivative ∂ (Theorem 22); and because base change coincides with container substitution on the linear layer (Lemma 7), this is at the same time the statement that ∂ is the unique nontrivial representable first-order tangent structure on that layer of finite-support polynomial functors. The proof ports Lanfranchi–Lemay's affine-scheme classification in both halves, and in both the rig $\mathbb{N}$ makes the port *cleaner*: freeness is automatic, so the rigidity needs no PID (Lemma 9); and the vertical bundle is a direct sum of free summands, so the converse needs no subtraction (Lemma 17, Theorem 18). Two technical observations carry the port: on the linear layer the base-change tensor $\otimes D$, the substitution tensor $\triangleleft$ and the Dirichlet tensor all coincide (Lemma 7), and free $\mathbb{N}$-modules are closed under direct sum with rank additive over it, so ranks add and never subtract. The arithmetic of the finite case is Lean-certified with no classical axioms. Finiteness turns out *not* to be load-bearing: although the countable part $\mathbb{N}\cdot y$ is abstractly

solid (an abstract isomorphism $M \otimes M \cong M$ exists, by the Cantor pairing), abstract solidity is strictly weaker than underlying a tangent structure at infinite rank — the flip and lift-coassociativity axioms force the vertical lift to be the diagonal, excluding every rank ≥ 2 (Theorem 13). So $\mathbb{N} \cdot y$ is a phantom, not a third structure, and the uniqueness holds at every rank; the earlier draft's "finiteness is sharp" is withdrawn (Remark 15).

Precedent. The hope that polynomial functors carry a differential structure is Cockett's. His unpublished FMCS 2012 talk *Can you Differentiate a Polynomial?* [6] conjectured that they form a Cartesian differential, hence a tangent, category — the thesis the companion note [11] later proved. The 2012 slides, though, are explicit that this was conjecture: the differentiation there is defined only on "separable" polynomials, closure under composition is asserted rather than proved, and the chain rule is named as the hard open point. (The slides even record, as an aside, that this is "a differential category in which negation is unnatural" — the same absence of additive inverses that we have just turned to advantage, since it is exactly what makes the vertical-bundle coproduct, rather than a short exact sequence, the right tool.) No uniqueness statement appears in 2012. The present note supplies it, and so answers Cockett's question in its strongest form: not only *can* you differentiate a polynomial, but — among representable Weil-base-change tangent structures — differentiation is the *only* nontrivial way to do so. We flag one caveat, due to Cruttwell via [10]: combinatorial species — the setting closest to Cockett's — are most naturally a tangent *bicategory*, with species as 1-morphisms, which is structurally different from the SPoly one-category treated here; our uniqueness is a statement about the latter.

Sequel. A rigid tangent structure is the entry condition for the Cartan-calculus / Lie-algebroid program of Aintablian and Blohmann [3], which asks when a tangent category supports a scalar "ring object" R making each TX an R-module compatibly with the tangent axioms, and thereby upgrades a tangent structure to a Lie-algebroid structure with a full Cartan calculus. That the linear layer of SPoly now has a *unique* nontrivial representable first-order tangent structure makes the question well-posed: is there such an R for $(\mathsf{SPoly}, \partial)$? The negation remark above is a warning sign — a scalar ring object requires additive inverses, which $\mathbb{N}$ does not have — and determining exactly how much of the Cartan calculus survives over a rig is the natural next problem. We leave it, together with higher Weil algebras $\mathbb{N}[\varepsilon]/\varepsilon^k$ ($k \geq 3$) and iterated tangents (first-order out of scope here, as for [1, 9, 5]), to future work.

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